

Education

University of Santiago de Cali

Master's in Computer Systems

Graduated from the University of Santiago de Cali (July 2026). My research focused on developing automated systems for SMEs (small and medium size enterprises), streamlining manual processes and integrating with existing workflows.

Systems Engineering - License 171122-0620819 VLL

Graduated from the University of Santiago de Cali (November 2021). Participated in university-sponsored robotics and programming events, working on low-level robotic programming and coding challenges. My contributions later inspired projects in machine learning for differential equations and computer vision for biometric access control.

Thesis

"Analysis of the benefits and risks of using code coverage", earned the highest grade (5.0/5.0) from three judges, my advisor, and the dean. Registered with the university library.

Professional Objective

My goal is to keep **turning ideas into high-impact digital platforms**, seamlessly **integrating web and mobile experiences**. I am driven by process **automation as an engine for continuous improvement**.

I seek an environment where I can apply my ability to **analyze and solve business problems** with a logical and creative mindset, **working collaboratively** to build robust, scalable solutions that **deliver tangible value to the organization** and its customers.

Experience

Full Stack Developer | Ministry of Sport (Colombia)

Mar 2020 – Sep 2020 (Parallel project)

- Developed a full-stack application for registration and management of community and intercollegiate games, serving 500+ schools and 10,000+ athletes.
- Reduced registration processing time from 3 days to 2 hours by automating workflows
- Stack: React, AWS Lambda, API Gateway, DynamoDB, AWS Cognito, AWS SES.



Software Developer | Public Transport Inspection Module

Jun 2020 – Dec 2020 (Parallel project)

- Developed backend microservices for compliance/non-compliance reports, 472 notifications (notifications of documentation, traffic reports and administrative acts)
- Implemented role-based access control (RBAC) for inspectors, administrators, and external auditors.
- Reduced manual paperwork by automating report generation and notification workflows.
- Stack: C# (.NET Core), Azure Functions, Azure SQL, SignalR, Angular.

Mobile Developer | Banco W – "Powwi" App

Sep 2020 – Apr 2021

- Built a mobile app for banking users to withdraw at ATMs, pay utilities, buy streaming pings, and send money peer-to-peer.
- Integrated with Banco W's existing REST APIs and implemented secure JWT authentication with token refresh.
- Published on App Store (iOS) and Play Store (Android) using React Native/Expo.
- Stack: React Native, Expo, JWT, FCM, REST APIs.

Backend Engineer | PSE Payment Gateway (Legacy Migration)

Apr 2021 – Dec 2022

- Rewrote a PSE payment gateway from scratch for a legacy PHP/JS app.
- Achieved official PSE certification on first attempt; service now processes 5K+ monthly transactions.
- Tech: TypeScript, NestJS.

Backend Developer | Sideral Marketplace

Dec 2022 – May 2023

- Built a multi-tenant marketplace that aggregates inventory from multiple pharmacy instances (each running the same Sideral platform, originally PHP/Laravel).
- Designed and implemented microservices for real-time inventory querying across tenants, enabling centralized product search (similar to Amazon).
- Stack: Bun (runtime), NestJS (backend framework), Drizzle (ORM), PostgreSQL





Mobile & Web Developer | Lotto/Sports (Puerto Rico & USA)

May 2023 – Oct 2023 (Parallel project)

- Developed a cross-platform mobile app (React Native/Expo) and a complementary web dashboard for lottery purchases and sports betting.
- Integrated real-time odds updates via WebSockets, reducing latency compared to polling.
- Implemented secure payment gateways (Stripe, PayPal) and compliant age-verification flows for US and Puerto Rico markets.
- Distributed via App Store and Play Store (React Native/Expo).
- Stack: React Native, Expo, React (web), Node.js backend, WebSockets, Stripe, PayPal.

Software Engineer | Ayesa & Enel Colombia

Jul 2023 – Dec 2023

- Built an e-invoicing module (React-PDF) with customer info, barcode, and legal
- Redesigned the platform's layout and UI, introducing a modern, responsive, and more intuitive interface – resulting in better and modern look and feel.
- Managed Codensa card data and Colpatria portfolio purchases.
- Stack: Java/Kotlin backend, React frontend, MongoDB, React-PDF.

Independent Full Stack Developer | Residuos Ambientales SAS

Dec 2023 – Present 2026

- Built a complete platform from scratch for waste collection management, including:
 - Public website for rate calculation and company info.
 - Admin portal covering: quote form, document verification, contract generation & signing, request approval, service conversion, route scheduling, operator manifests (with cargo reports), invoice generation, and certificate delivery.
- Orchestrated background jobs with Inngest.
- Stack: Next.js, Supabase, Bun, Vercel deployments.



Independent Mobile Developer | Fedelta Mall SAS – E-commerce App

Apr 2024 – Dec 2025

- Developed a React Native e-commerce app with an integrated admin dashboard.
- Features: multi-store management, product upload/publishing, Woocommerce (Woompi) payment gateway integration, in-app reporting system, advertising management.
- Published to App Store and Play Store.
- Stack: React Native, Node.js backend, dashboard built with React/Next.js.

Languages, Databases & Framework

Languages & versions

- C++ using standard 17, 14, 11 and 98
- C# using .Net, .Net Core
- Java using Oracle JDK 21, 17, 11 and 8
- Javascript using ESNext, ES2022
- Typescript using v5 and v6
- Python using Version 3.7 and 3.11
- C# using .Net, .Net Core

Framework & Library

- React, Next.js, Tanstack Starter
- React Native, Expo
- Angular, Php, Laravel
- NestJS, .NET Core
- React-PDF

Testing & Quality

- Unit/Integration/System tests
- Code coverage (gcov/lcov)
- Valgrind
- JUnit, JTest, JMock, Mockito

Databases

- PostgresSQL (Supabase)
- MongoDB 4.4.3
- MySQL 8 & 5.7
- DynamoDB
- SQLite & LibSQL

Cloud & Services

- AWS (Lambda, API Gateway, SES)
- Azure (Functions, SQL)
- Vercel, Inngest, Bun, Node
- Stripe, WebSocket, FCM

Other

- Memory leak detection
- Binary analysis
- Metaprogramming
- CI/CD implementation
- Refactoring large codebases
- Microservices architecture

